CRISIS INTERVENTION FOR PARAMEDICS

Unit 5: Psychotic illness

Marilyn Niffin BSc, ACP
Professor Georgian College
Winter 2020, 2021, 2023

Learning Objectives

- Identify common psychotic disorders
- Differentiate between various psychotic disorders
- Describe the assessment principles for patients suffering from psychosis
- Identify management principles for patients suffering from psychosis

Types of Mental Illness that include Psychosis

- Schizophrenia Spectrum Disorder*
  - *Schizoaffective Disorder*
  - *Delusional disorder*
  - *Schizophreniform disorder*
  - *Brief psychotic Disorder*
  - *Schizotypal Personality Disorder*
- Major Depressive Disorder
- Bipolar Disorder
- Drug induced psychosis
- Depression with psychotic features

What is Psychosis

- Conditions affecting the mind when people have trouble distinguishing what is real and what is not
- 3 of every 100 people will experience a psychotic episode (camh, 2019)
- Affects men and women equally across all cultural and socioeconomic backgrounds
  - *Usually appears first in the late teens or early 20's*
  - *Tends to affect women at a later age than men*
  - *Women tend to respond better to treatment than men, but are more likely to relapse (may be related to hormonal changes)*
- Includes hallucinations and delusions
  - *Hallucinations; Visual, tactile, auditory, olfactory*
  - *Delusions; Erotomanic, Grandiose, Jealousy, Persecution, Somatic, nihilistic, referential*

Psychotic Illness

- Causes
  - *Genetic Predisposition*
  - *Mental health disorders*
  - *Drug Use/Abuse*
  - *Medical Conditions*

Genetic Predisposition

- Twin Studies – rates of concordance are higher for identical twins than fraternal twins suggesting a strong genetic component
- Adoptive studies
- Gene mutations

Drug Use/Abuse

- Hallucinogenic drugs – can produce features of psychotic disorders in otherwise healthy individuals
- Other drugs associated with higher risk of developing psychotic disorders
  - *Psychostimulants (ephedra, cocaine, amphetamines)*
  - *Inhalants (nitrous oxide)*
  - *Cannabis*
  - *LSD, PCP, Ecstasy*
  - *Ketamine*
  - *Alcohol*

Medical Conditions

- Renal failure
- Electrolyte imbalances
- Head injuries
- Degenerative diseases (Alzheimer’s, Parkinson’s)
- Delirium
- Infection (syphilis, HIV)
- Thyroid disease
- Vitamin deficiencies (B12)
- Insulinoma, pheochromocytoma
- Metabolic diseases

Research

- Neurological and neuroimaging studies that look at the brains of patients with psychotic disorders show:
  - *Less activity in the prefrontal cortex (compared to control subjects – no psychotic disorder)*
  - *Larger ventricles in schizophrenic patients (vs. control patients)*
  - *Stronger activity in the right hemisphere during language and interpretation*

Schizophrenia Spectrum Disorder

- A chronic mental health disorder that exists on a spectrum
- A disorder that interferes with a person's perception of reality
- Characterized by relapsing episodes of psychosis
  - *Includes hallucinations, delusions, disordered thinking and speech, social withdrawal, decreased emotional expression, lack of motivation, paranoia, catatonia, agitation or aggression*
- Strong genetic component
- More common in low birth weight babies (abnormal fetal brain development)
- Slightly more common in men
- Typically starts in late teens and early 20's; later onset for women**
  - *Most common age range (21-25 for males and 25-30 for women)*
- Increased risk with Cannabis use (in teens and early 20's) and heavy alcohol use
- ***Early diagnosis significantly improves prognosis and outcomes***

Schizophrenia Spectrum Disorder

- Some early onset symptoms:
  - *Saying things that don't make sense to those around you*
  - *Unusual behaviour*
  - *Bizarre thoughts, ideas or beliefs*
  - *Decline in academic performance*
  - *Personality changes/mood shifts/withdrawal*
  - *Decline in personal hygiene*
  - *Persistent confusion*
  - *Visual or auditory hallucinations*
  - *Mixing fantasies (TV stories or dreams) with reality*
- Due to the condition many people with schizophrenia are not able to establish an awareness with the symptoms they are having (anosognosia)

Schizophrenia

Diagnostic criteria

- Symptoms (at least two) present for at least one month and last for at least 6 months and have a severe impact on person's social, educational or occupational functioning/abilities
  - *At least 1 symptom must be delusions, hallucinations, or disorganized speech*
- Variations include (Schizophrenia spectrum disorders):
  - ***Schizophreniform disorder:***
  - ***Schizoaffective disorder;***
  - ***Delusional Disorder:***
  - ***Brief psychotic disorder***
  - ***Schizotypal personality Disorder***

Schizophrenia

- Type I schizophrenia
  - *Positive-syndrome*
- Negative syndrome
  - *Type II schizophrenia*

Type I & II

- Positive Syndrome
  - *Hallucinations*
  - *Delusions*
  - *Derailment of Assoc.*
  - *Bizarre Behavior*
  - *Minimal Cognitive Impairment*
  - *Sudden Onset*
  - *Variable Course*
- TYPE I.----------
- The above plus
  - *Good Med Response*
  - *Limbic System Abnormality*
  - *Normal Brain Ventricles*

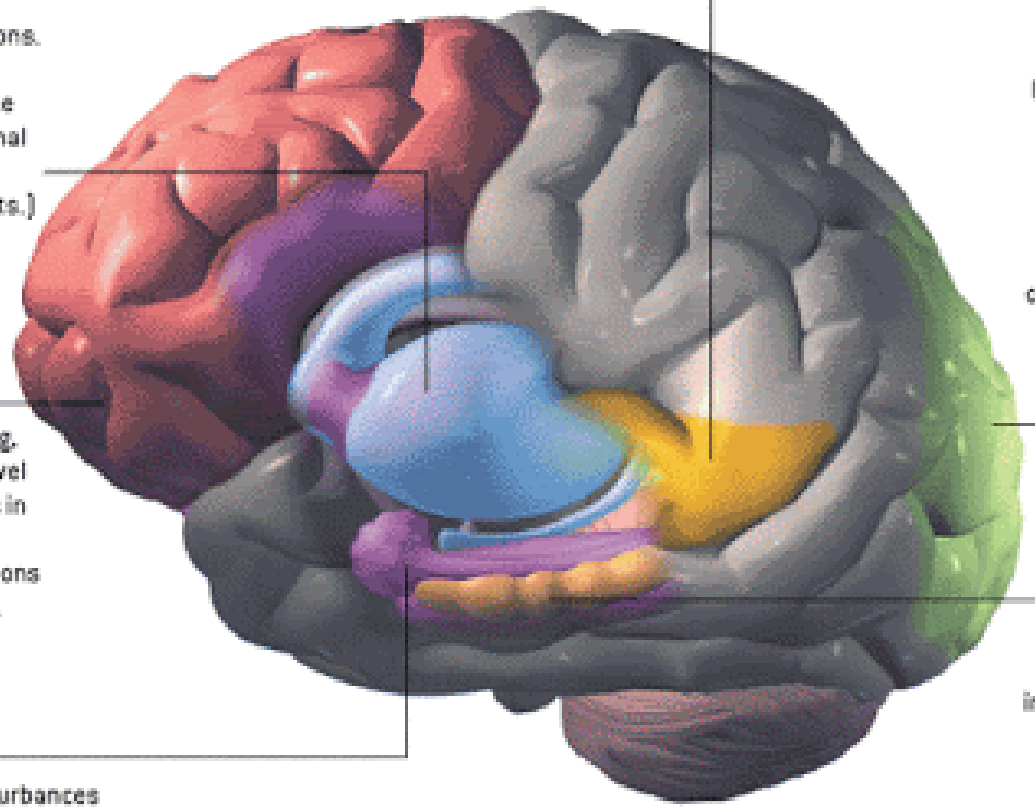
- Negative Syndrome
  - *Emotional Flattening*
  - *Poverty of Speech*
  - *Asociality (social withdrawal)*
  - *Apathy (difficulty planning or executing*
  - *Significant Cognitive Impairment*
  - *Insidious Onset*
  - *Chronic Course*
- Type II.----------
- Plus
  - *Uncertain Med Response*
  - *Frontal Lobe Abnormality*
  - *Enlarged Brain Ventricles*

THE BRAIN IN SCHIZOPHRENIA

MANY BRAIN REGIONS and systems operate abnormally in schizophrenia, including those highlighted below. Imbalances in the neurotransmitter dopamine were once thought to be the prime cause of schizophrenia. But new findings suggest that impoverished signaling by the more pervasive neurotransmitter glutamate—or, more specifically, by one of glutamate's key targets on neurons (the NMDA receptor)—better explains the wide range of symptoms in this disorder.

BASAL GANGLIA
Involved in movement and emotions and in integrating sensory information. Abnormal functioning in schizophrenia is thought to contribute to paranoia and hallucinations. (Excessive blockade of dopamine receptors in the basal ganglia by traditional antipsychotic medicines leads to motor side effects.)

AUDITORY SYSTEM
Enables humans to hear and understand speech. In schizophrenia, overactivity of the speech area (called Wernicke's area) can create auditory hallucinations—the illusion that internally generated thoughts are real voices coming from the outside.



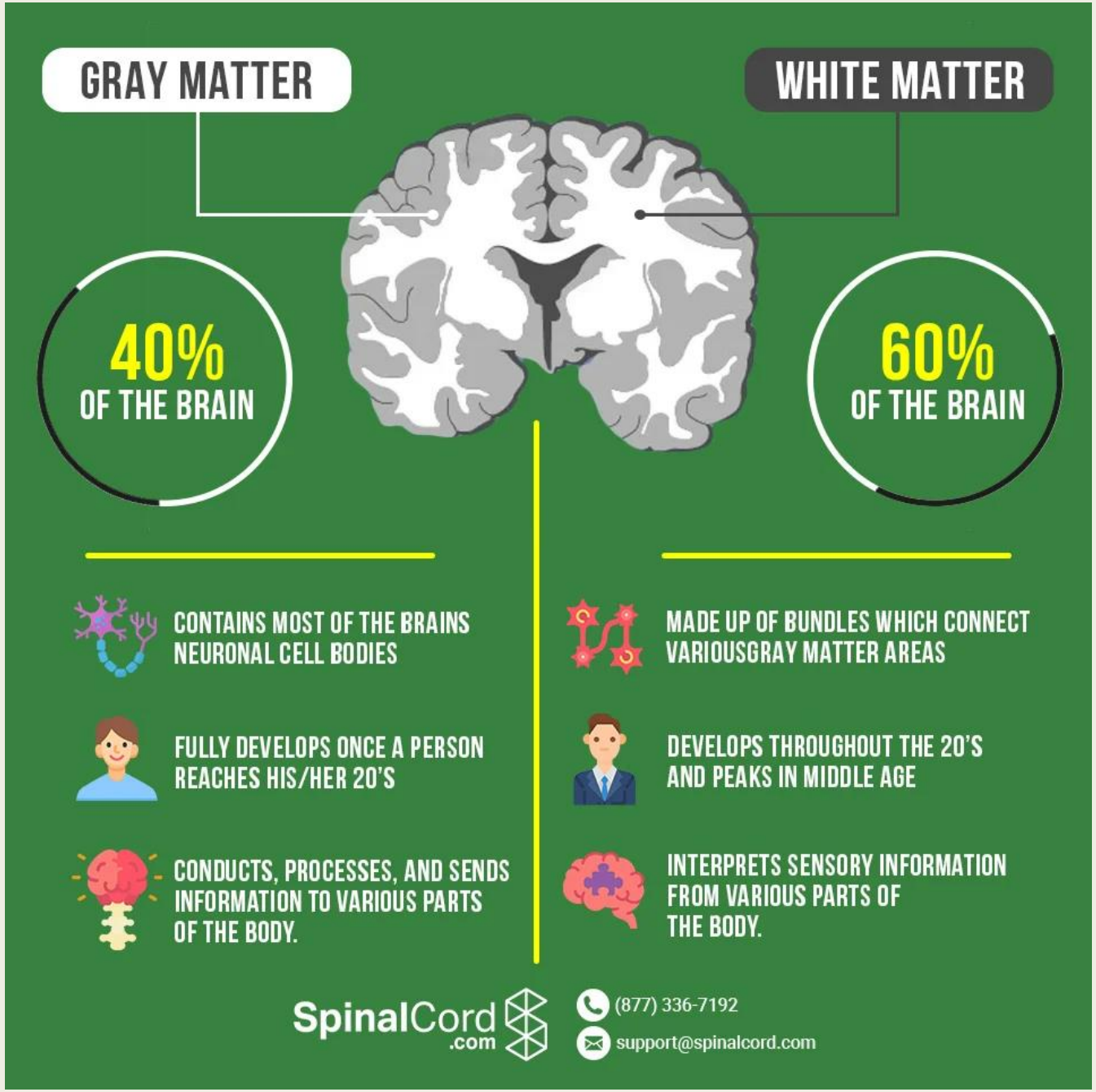
OCCIPITAL LOBE
Processes information about the visual world. People with schizophrenia rarely have full-blown visual hallucinations, but disturbances in this area contribute to such difficulties as interpreting complex images, recognizing motion, and reading emotions on others' faces.

FRONTAL LOBE
Critical to problem solving, insight and other high-level reasoning. Perturbations in schizophrenia lead to difficulty in planning actions and organizing thoughts.

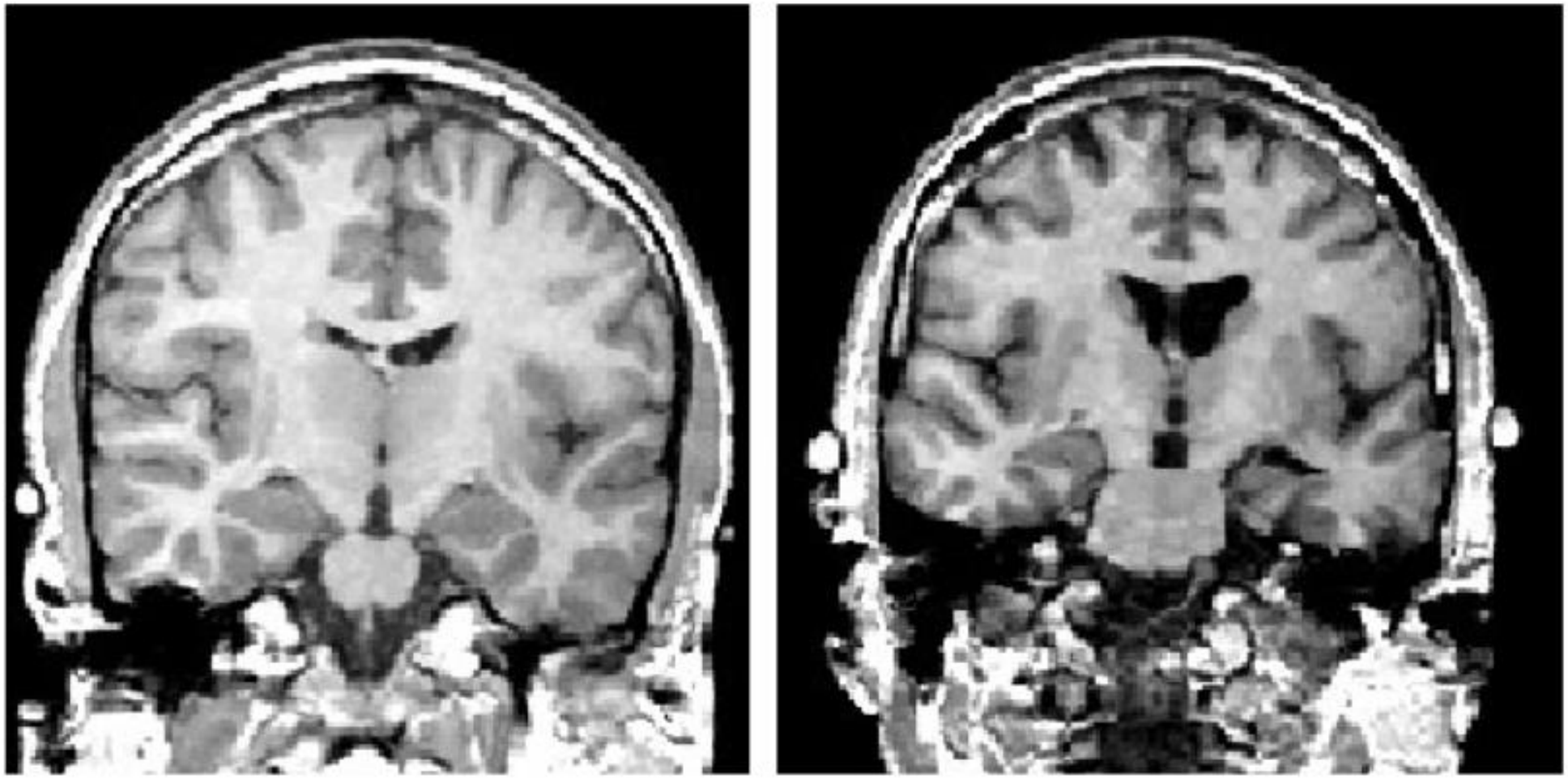
HIPPOCAMPUS
Mediates learning and memory formation, intertwined functions that are impaired in schizophrenia.

LIMBIC SYSTEM
Involved in emotion. Disturbances are thought to contribute to the agitation frequently seen in schizophrenia.

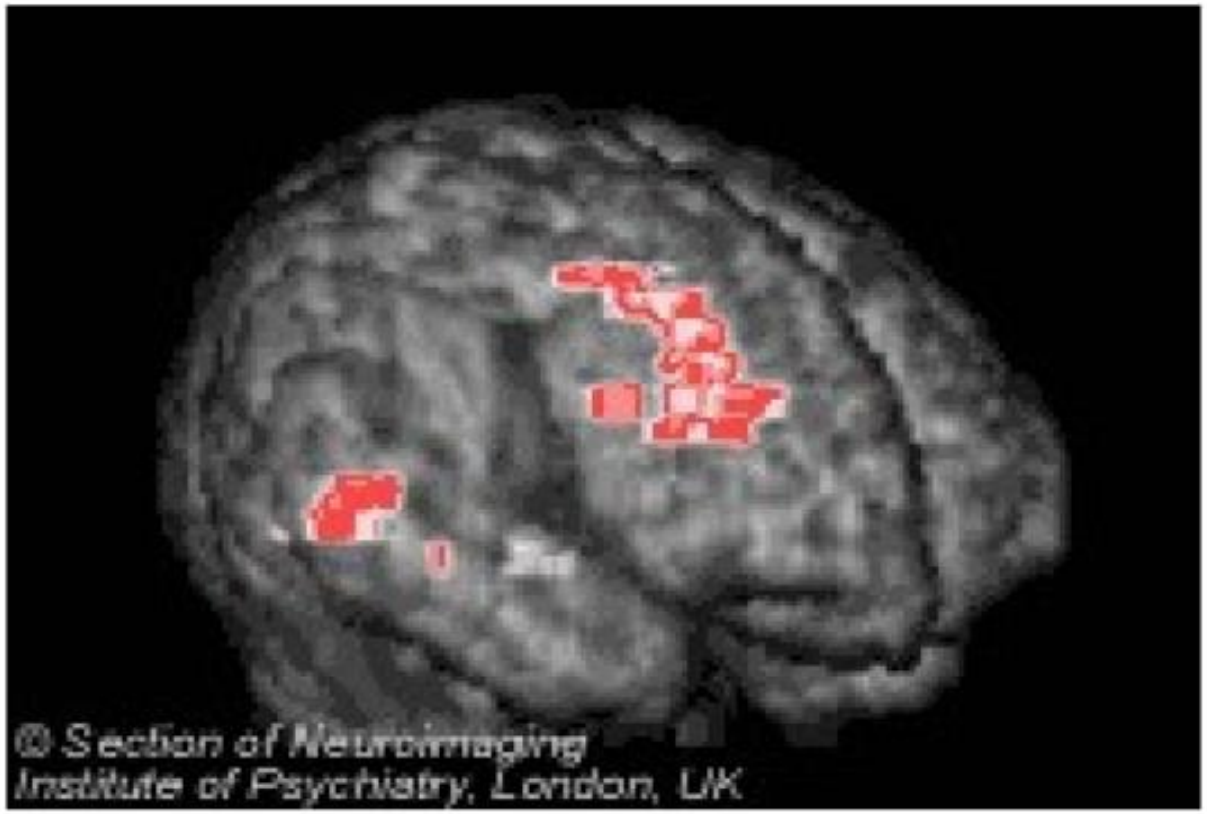
ALFRED T. KAMAJIAN



Normal Brain Development



Left: Brain of a healthy control subject
Right: Brain of a schizophrenic patient



fMRI scan of Schizophrenic patient having an auditory hallucination

Schizophrenia - Treatment

- Antipsychotic medication
- +/- Mood stabilizers, antidepressants, sedatives and medications to help with the side effects of the antipsychotics
- ECT, transcranial magnetic stimulation
- Psychosocial therapies/support; psychoeducation, CBT, family therapy, diet, exercise, structured daily routines
- Hospitalization – during exacerbation of symptoms and periods when patients are not coping/ unable to care for themselves; days to weeks

Cleveland Clinic
WARNING
SIGNS OF
SCHIZOPHRENIA

Schizoaffective Disorder

- Essentially schizophrenia with the presence of a mood disorder
  - *Symptoms of schizophrenia accompanied by either depression or mania*
  - *The symptoms of both (schizophrenia and mania or depression) are present at the same time AND*
  - *The symptoms of schizophrenia are present outside of the depression or mania*

Schizophreniform Disorder

- Very similar to schizophrenia however this is differentiated by time
  - *a person has some psychotic symptoms for more than one month and less than six months.*

Delusional Disorder

- characterized by the presence of **non-bizarre** delusions in the absence of other mood or psychotic symptoms
- Defined as false beliefs based on incorrect inference about external reality that persist despite the evidence to the contrary
- Non-bizarre: delusions about situations that could occur in real life, such as being followed, being loved, having an infection, and being deceived by one's spouse
- Bizarre delusions = more severe types of psychotic illnesses and "are clearly implausible, not understandable

Brief psychotic disorder

- – sudden onset of psychotic symptoms (hallucinations, delusions, disorganized thinking or other unusual behaviour and confusion)
- that last < 1 month
- Rare and often brought on by extreme stress

Schizotypal personality Disorder

- These persons are described as strange or eccentric
- Exhibit odd behaviours
- Are anxious in social situations
- Tend NOT to have delusions or hallucinations

Major Depressive disorder

- Also called clinical depression, is a complex mood disorder
  - *Seasonal affective disorder (SAD)*
    - Most often occurs during fall and winter; associated with fewer daylight hours
  - *Perinatal depression (formerly Postpartum depression)*
  - *Psychosis – depression is so severe that affected persons lose touch with reality and can experience hallucinations*

Delusional Disorder

- Bizarre delusions = more severe types of psychotic illnesses (eg, schizophrenia) and "are clearly implausible, not understandable
- Cultural caution: some cultures have widely accepted beliefs that may be considered delusional in other cultures
- patients with delusional disorder do not have good insight into their pathological experiences

Psychosis - Treatment methods

- Non-pharmacological treatments
  - *Psychosurgery (e.g. lobotomy)*
  - *Electro-convulsive therapy (ECT)*
  - *Psychotherapy (CBT)*
- Anti-psychotic medications
  - *Dopamine receptor blocking agents (D2)*
    - Strong dopamine blockers also block the dopamine required for muscle movement (tardive dyskinesia)
    - Other side effects: dizziness, sedations, difficulty urinating, constipation, sexual dysfunction
    - **Chlorpromazine (Thorazine),** thiothixene (Navane), **haloperidol (Haldol),** loxapine (Loxatane) molindone (Moban) pimozide (Orap)
  - *Antidepressants*
  - *Mood stabilizers*

Treatment Methods

- Atypical antipsychotic medications
  - *Weaker D2 blockers with serotonin blocking effects*
  - *Clozapine, risperidone (Risperdal), olanzapine (Zyprexa), quetiapine (Seroquel), aripiprazole (Abilify)*
  - *Side Effects; weight gain, extrapyramidal effects (slowed movements shuffling gait), sedation*

Management

- Attempt to calm your patient – determine their needs
- Try to orient them to the environment at hand
- NEVER lie to your patient
- Attempt to build a rapport and trust with your patient.
- Offer choices when appropriate
- Maintain dignity

Management

- Be aware of the potential for violence
- Assess for history of violent or aggressive acts (lack of history dose not rule out the risk of violence)
- Maintain egress –never let the patient stand between you and the exit
- Use family members to gain information (as appropriate). If they appear to be exacerbating the situation, ask them to leave
- Don’t dismiss red flags
- Ask for police when in doubt (if you feel you may be in danger, it is ok to wait for police to enter the scene. Yours and your partners safety always comes first)

- Resources
- https://www.jems.com/2014/07/10/evaluation-and-management-psychiatric-em/
- https://www.camh.ca/en/health-info/mental-illness-and-addiction-index/psychosis
- https://www.psychiatry.org/psychiatrists/practice/dsm/educational-resources/dsm-5-fact-sheets